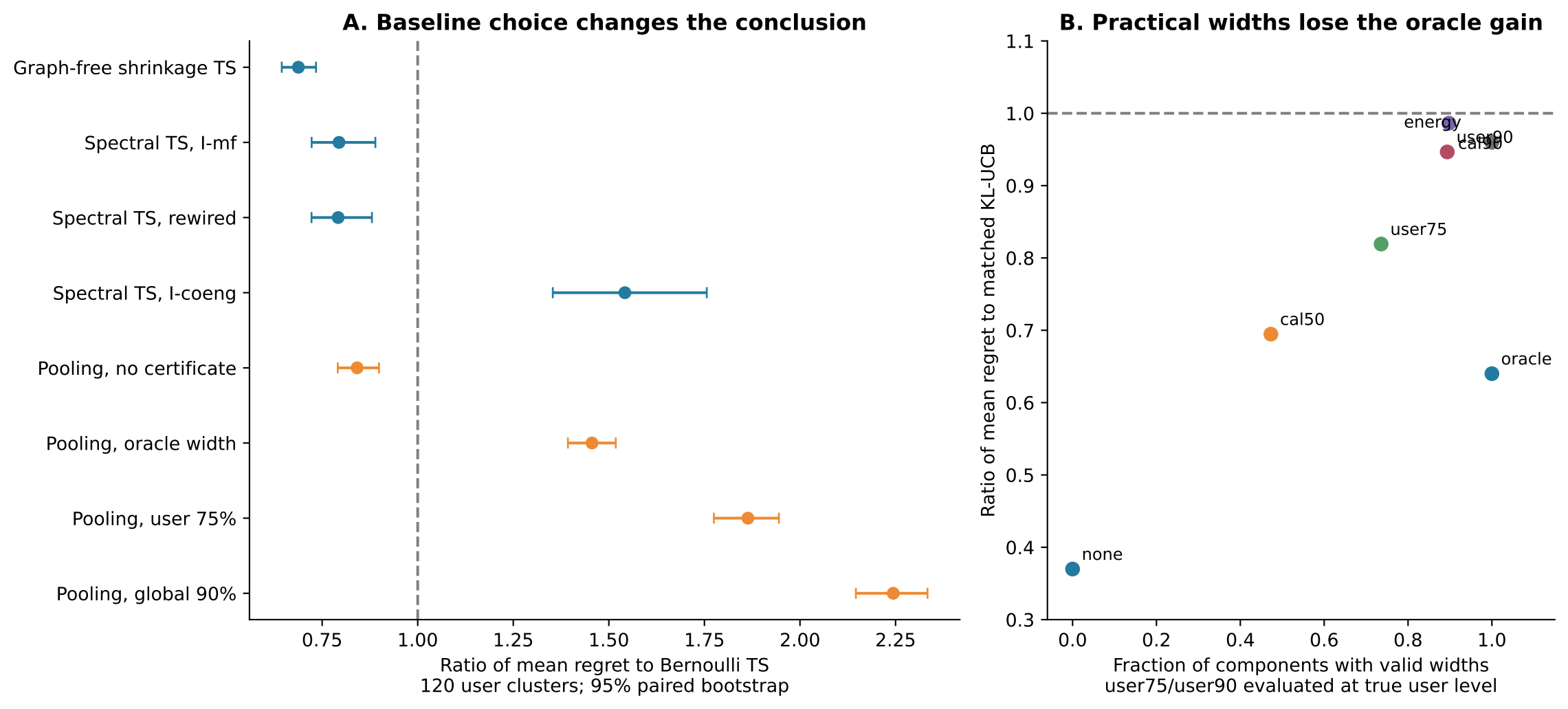
A. Baseline choice changes the conclusion
Graph-free shrinkage TS
Spectral TS, I-mf
Spectral TS, rewired
Spectral TS, I-coeng
Pooling, no certificate
Pooling, oracle width
Pooling, user 75%
Pooling, global 90%
0.75
1.00
1.25
1.50
1.75
2.00
2.25
Ratio of mean regret to Bernoulli TS
120 user clusters; 95% paired bootstrap
B. Practical widths lose the oracle gain
Ratio of mean regret to matched KL-UCB
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
energy
user90
cal90
user75
cal50
oracle
none
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of components with valid widths
user75/user90 evaluated at true user level